Inspect and clean the data

Project: Observed Coverage and Response-Time Monitor **Student:** Yoojin Hwang

Fill in the bracketed fields. For each item, choose a status and give evidence. Explain every Not applicable or Unresolved response. Add space as needed.

1 State the project vision and question

Vision / decision: A Community Notes Product Manager must decide where additional note-writing or rating effort should be directed to improve coverage and reduce response time while maintaining broad agreement.

Business question: Which request feeds have lower observed note coverage, and how long do notes take to reach their first non-NMR status?

Main measures: Observed coverage, coverage gap, response time to first non-NMR status, and observed helpfulness.

Status: Done

Evidence or explanation: The project is an Observed Coverage and Response-Time Monitor. The measures describe the downloaded data and help identify areas for further review.

2 Define one row

Notes: One row represents one Community Note. Unique identifier: noteId.

Ratings: One row represents one participant's rating of one note. Unique identifier: noteId + raterParticipantId.

Note Status History: One row represents one note's status-history record. Unique identifier: noteId.

Note Requests: One row represents one post/tweet with a note request record. Unique identifier: tweetId.

Status: Done

Evidence or explanation: Full-file checks found zero duplicate keys in all four datasets. The 27,038,662 Ratings records also had zero missing noteId or raterParticipantId values. See "Define One Row" and "Ratings Full-file Check" in data_cleaning_organized.ipynb.

3 Preserve the source

Data source: X Community Notes Public Data

Download date: October 2, 2026

Original files: notes-00002.tsv, ratings-00006.tsv, noteStatusHistory-00000.tsv, and batSignals-00000.tsv

Source preservation: The original TSV files are unchanged. They are loaded with pandas.read_csv() using tab separators. Ratings are read in chunks for full-file verification.

Status: Done

Evidence or explanation: See "Load Data" and "Preserve the Source" in data_cleaning_organized.ipynb.

4 Inspect columns and types

Incorrect types and conversions:

Data types were checked for all four datasets. In the Ratings full-file scan, noteId, raterParticipantId, and suggestion were loaded as strings. This kept IDs as text and avoided mixed-type inference for suggestion. Each Ratings column had the same dtype across all chunks.

Status produced mixed-type import warnings for lockedStatus and currentMultiGroupStatus. Follow-up checks found that all nonmissing values in both columns were strings. No further conversion was needed. Numeric columns with missing values were kept as loaded.

Failed conversions: 0. String types were specified during loading; no later value conversion was applied.

Status: Done

Evidence or explanation: See "Inspect Columns and Types," "Status Type Checks," and "Ratings Full-file Check" in data_cleaning_organized.ipynb.

5 Standardize text

Before/after examples:

No text standardization was needed for the inspected categories. HELPFUL stayed HELPFUL, NOT_HELPFUL stayed NOT_HELPFUL, and NEEDS_MORE_RATINGS stayed NEEDS_MORE_RATINGS.

The full Ratings file contained HELPFUL, SOMEWHAT_HELPFUL, and NOT_HELPFUL in helpfulnessLevel. ratingSourceBucketed contained DEFAULT and POPULATION_SAMPLED.

Meaningful differences and IDs:

Distinct status categories were kept separate. Free-text fields such as summary, sourceLinks, suggestion, and suggestions were unchanged. Participant IDs were not trimmed or recoded.

Status: Done

Evidence or explanation: See "Standardize Text Values," "Status Type Checks," and "Ratings Full-file Check" in data_cleaning_organized.ipynb.

6 Investigate duplicates

Exact duplicate rows:

Notes, Note Status History, and Note Requests had zero exact duplicate rows in full-file checks. Ratings had zero exact duplicate rows because its complete noteId + raterParticipantId key was unique across all 27,038,662 records.

Repeated ID:

Notes had 5,234 repeated tweetId occurrences beyond the first occurrence. These were kept because multiple notes can refer to one tweet. In the 1,000-row Ratings sample, noteId repeated 991 times; this is a sample count, not a full-file total. Multiple participants can rate one note.

The full-file duplicate count was zero for Ratings noteId + raterParticipantId, Notes noteId, Status noteId, and Requests tweetId.

Kept/removed: All records were kept. No duplicate rows were removed.

Status: Done

Evidence or explanation: See "Handle Duplicates" and "Ratings Full-file Check" in data_cleaning_organized.ipynb. The Ratings key check included duplicates across chunk boundaries.

7 Handle missing data

Missing counts before and after cleaning:

Before = After for every column below. No missing values were filled, and no records were dropped.

Notes:

- classification: 1
- believable: 13,649
- harmful: 13,649
- validationDifficulty: 13,649

Ratings:

- suggestion: 27,038,157

Note Status History:

- timestampMillisOfFirstNonNMRStatus: 2,767,657
- firstNonNMRStatus: 2,767,657
- timestampMillisOfLatestNonNMRStatus: 2,767,657
- mostRecentNonNMRStatus: 2,767,657
- timestampMillisOfStatusLock: 240,179
- lockedStatus: 240,179
- timestampMillisOfRetroLock: 3,342,306
- currentCoreStatus: 706,547
- currentExpansionStatus: 183,861
- currentGroupStatus: 2,289,825
- currentModelingGroup: 2,289,825
- timestampMillisOfNmrDueToMinStableCrhTime: 1,170,025
- currentMultiGroupStatus: 3,280,453
- currentModelingMultiGroup: 3,280,453
- timestampMillisOfFirstNmrDueToMinStableCrhTime: 3,112,207
- timestampMillisAbovePcrhThreshold: 3,317,870

Note Requests:

- sourceLinks: 372,919

All other columns had zero missing values in the full-file checks.

Handling and reasons:

Missing values were kept because there was no reliable basis for filling them. The missing Notes classification was not inferred. Status fields and timestamps were not replaced with invented values or zeros. Suggestion and source-link fields were left missing.

Status: Done

Evidence or explanation: See "Handle Missing Values" in data_cleaning_organized.ipynb. All four full datasets were checked.

Validate the results

Use the same statuses. Give evidence for Done. For Unresolved, explain how the issue affects your analysis.

8 Check suspicious values

Suspicious values: Six Notes records had tweetId = -1. All six had isCollaborativeNote = 1.

Handling: These records were kept and identified for review. The correct tweet IDs were unavailable, so they were not changed or removed. The value -1 was treated as unusual, not automatically as an error.

Status: Done

Evidence or explanation: See "Check Suspicious Values" in data_cleaning_organized.ipynb.

9 Verify merges

Merge 1 - Note Requests to Notes

Key: tweetId

Expected relationship: One request/tweet may match multiple notes.

Requests before merge: 487,851
Notes before merge: 13,649
Rows after merge: 489,809
Unmatched requests: 484,056

The increase in rows is consistent with multiple notes matching one tweetId.

Merge 2 - Notes to Note Status History

Key: noteId

Expected relationship: Each note should match at most one status-history record.

Notes before merge: 13,649
Status History before merge: 3,343,818
Rows after merge: 13,649
Unmatched notes: 3,044

No unexpected extra rows were created.

Status: Done

Evidence or explanation: See "Verify Merges" in data_cleaning_organized.ipynb.

10 Reconcile changes

Starting and final row counts:

- Notes: 13,649 → 13,649
- Ratings: 27,038,662 → 27,038,662
- Note Status History: 3,343,818 → 3,343,818
- Note Requests: 487,851 → 487,851

No source records were added or removed. Missing and unusual values were kept, and no duplicate rows required removal.

Merge reconciliation: Requests-to-Notes increased from 487,851 to 489,809 rows because one tweetId can match multiple notes. Notes-to-Status remained at 13,649 rows.

Important totals: The original row counts above were preserved.

Status: Done

Evidence or explanation: See "Reconcile Changes" and "Verify Merges" in data_cleaning_organized.ipynb.

11 Check calculations

Observed helpfulness:

Numerator: 16,820,314 HELPFUL ratings.
Denominator: 27,038,662 ratings with a valid helpfulnessLevel.
Result: 62.21%.

Valid values were HELPFUL, SOMEWHAT_HELPFUL, and NOT_HELPFUL. No ratings were excluded for missing or invalid helpfulnessLevel. Missing suggestion values did not affect this calculation.

Response time to first non-NMR status:

Response time in hours = (timestampMillisOfFirstNonNMRStatus − Notes createdAtMillis) / 3,600,000.

Of 13,649 Notes, 552 had the timestamps needed for a nonnegative duration. Median response time was 6.50 hours; the additional mean check was 22.89 hours. The remaining 13,097 records had an unavailable required timestamp and were excluded, not assigned zero. No negative durations were found. The cutoff check retained the same 552 records.

Observed coverage and coverage gap:

Each feed was checked separately using all eligible dates through October 2, 2026, 23:59:59.999 UTC. This is the verification cutoff, not a confirmed final dashboard setting. Eligibility values of -1 and invalid post IDs were excluded.

Denominator: distinct eligible tweetId values. Numerator: eligible posts with at least one observed note created after eligibility and by the cutoff. Each post was counted once. Coverage gap = denominator − numerator.

Standard Note Request feed: 550 / 28,597 = 1.92%; gap: 28,047 posts.

Small API feed: 556 / 28,964 = 1.92%; gap: 28,408 posts.

Large API feed: 2,092 / 168,583 = 1.24%; gap: 166,491 posts.

Extra-large API feed: 3,119 / 487,506 = 0.64%; gap: 484,387 posts.

Unmatched requests count as uncovered in the supplied files only. They do not prove that no note exists in other Notes files. Missing timestamps do not contribute to the eligible-post denominator or covered-post count.

Missing group keys:

These are overall calculations, repeated separately for each feed; no classification or time-period groups were used. Therefore no records were dropped for missing classification. The 552 included response-time records all had classification values.

Status: Done

Evidence or explanation: See “Ratings Full-file Check” and “Check Calculations” in data_cleaning_organized.ipynb. Numerators, denominators, exclusions, and results were checked.

12 Verify individual records

Five Record Checks

Five Notes records were selected. Before checking their values, the expected result and reason were recorded: each should remain unchanged because no cleaning rule required modification, deletion, or imputation.

The following checks compare original TSV values with the exported sample. The script defines the expected result and reason before loading actual values. All 24 source fields were compared for each record.

2014495336161829107

Original: tweetId=-1; classification=NOT_MISLEADING; isCollaborativeNote=1
Expected: all fields unchanged.
Actual: tweetId=-1; classification=NOT_MISLEADING; isCollaborativeNote=1
Match: Yes (all 24 source fields).

2014121093532098991

Original: tweetId=-1; classification=(missing); isCollaborativeNote=1
Expected: all fields unchanged.
Actual: tweetId=-1; classification=(missing); isCollaborativeNote=1
Match: Yes (all 24 source fields).

2077499944613937440

Original: tweetId=1414366007888650241; classification=NOT_MISLEADING; isCollaborativeNote=1
Expected: all fields unchanged.
Actual: tweetId=1414366007888650241; classification=NOT_MISLEADING; isCollaborativeNote=1
Match: Yes (all 24 source fields).

2035025007030473042

Original: tweetId=2004845222027821222; classification=NOT_MISLEADING; isCollaborativeNote=1
Expected: all fields unchanged.
Actual: tweetId=2004845222027821222; classification=NOT_MISLEADING; isCollaborativeNote=1
Match: Yes (all 24 source fields).

2022480596937978339

Original: tweetId=2021910752899240028; classification=NOT_MISLEADING; isCollaborativeNote=1
Expected: all fields unchanged.
Actual: tweetId=2021910752899240028; classification=NOT_MISLEADING; isCollaborativeNote=1
Match: Yes (all 24 source fields).

Status: Done

Evidence or explanation: See "Verify Individual Records" in data_cleaning_organized.ipynb and the printed checks from prepare_submission.py.

13 Test reproducibility

The kernel was restarted, and the notebook was run from the beginning using the original TSV files. All cells completed successfully. The workflow uses Python and pandas, with sqlite3, tempfile, and pathlib from the Python standard library. The source files remain unchanged.

Run instructions: Place the four TSV files at the paths in “Load Data,” select the Python environment with pandas, restart the kernel, and run all cells. The full Ratings file is scanned once.

Status: Done

Evidence or explanation: The updated notebook completed a top-to-bottom run. Imports, file paths, and the loading steps are shown in data_cleaning_organized.ipynb.

14 Document remaining limitations

Remaining limitations:

Some Note Requests do not match the supplied Notes, and some Notes do not match Status. Missing values remain because they could not be reliably filled. Coverage uses the supplied Notes file and a verification cutoff; the final dashboard feed and period still need to be chosen.

Effect on conclusions:

Coverage describes only notes observed in these files. Response-time results describe the 552 records with available timestamps. The helpfulness percentage describes ratings and does not measure broad agreement or reproduce the Community Notes scoring model. These results do not establish causation.

Additional information needed:

More complete, aligned files and information about unmatched records would help assess coverage. Post-level information and clearer meanings for missing or special values would help decide whether further cleaning is appropriate.

Status: Done

Evidence or explanation: See “Handle Missing Values,” “Verify Merges,” “Check Calculations,” and “Remaining Limitations” in data_cleaning_organized.ipynb.

Cleaning decision log

Decision 1

Rule: Preserve original TSV files and all source rows

Reason: Retain traceable source values; no justified row deletion

Check: Full row counts retained: Notes 13,649; Ratings 27,038,662; Status 3,343,818; Requests 487,851

Decision 2

Rule: Load Ratings noteId, raterParticipantId, suggestion as strings

Reason: Preserve IDs and avoid mixed-type inference for suggestion

Check: Types consistent across all chunks; key missing count 0

Decision 3

Rule: Keep missing values; do not fill with zero

Reason: Missing information is not a known zero or a negative outcome

Check: Before=after counts; Ratings suggestion missing 27,038,157

Decision 4

Rule: Keep tweetId=-1 Notes for review; exclude invalid post IDs from coverage

Reason: Cannot infer correct post IDs; unusual collaborative records should remain visible

Check: 6 records; all isCollaborativeNote=1; sample includes all 6

Decision 5

Rule: Keep repeated tweetId in Notes and noteId in Ratings

Reason: Multiple notes per post and multiple raters per note are valid

Check: Notes repeated tweetId 5,234; full Ratings composite-key duplicates 0

Decision 6

Rule: Exclude eligibility=-1; count distinct eligible posts per feed

Reason: -1 is not a real eligibility date; one post may have several notes

Check: Coverage denominators 28,597; 28,964; 168,583; 487,506

Decision 7

Rule: Compute response time only from available nonnegative timestamp pairs

Reason: Missing resolution times must not become zero hours

Check: 552 included; 13,097 missing; 0 negative; median 6.49916625 hours

Additional submission evidence

How I checked a Codex suggestion

Codex suggested counting duplicate rows within each Ratings chunk. I checked the method and found that it could miss duplicates across chunk boundaries. The final scan instead stores noteId + raterParticipantId keys in a temporary SQLite database with a unique key. Across all 27,038,662 rows, duplicate pairs and missing keys were both zero. This also rules out identical full rows. No rows were removed.

Data sample and selection

notes_cleaned_sample.csv contains 50 Notes rows and all 24 source columns. I included all six rows with tweetId=-1, including the row with missing classification, plus three normal verification IDs. I then selected the first remaining rows until the sample reached 50. Selected rows remain in their original order. This is a verification sample, not a random or representative sample.

prepare_submission.py recreates the sample without reading Ratings. It checks that all exported values and source-column names match the original, then prints five full-record comparisons. The script was run twice; the sample CSV and printed checks were identical across both runs.

Files for my GitHub branch

data_cleaning_organized.ipynb; prepare_submission.py; notes_cleaned_sample.csv; requirements.txt; README.md; this PDF.

GitHub submission details

Branch: YoojinHwnag

Branch URL: https://github.com/jiamaotian/misinformation-analysis/tree/YoojinHwnag

Commit ID: Pending commit and push of the prepared submission files.

The final commit ID must be entered here and in README.md after uploading. Instructor repository access must also be confirmed.